

Next Generation Carrier-based Strike Fighter Aircraft: F/A-67 Condor

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Request for Proposal

The United States Navy is seeking a replacement for the F/A-18E/F with improved performance and comparable unit cost. To maintain the number of aircraft in the inventory, Augustine's law must be broken. The new aircraft will have an initial production run of 500 units with 2035 operating capability and a 25-year service life. Existing engines must be used, all subsystems must be TRL 6 or higher, and internal carriage of ordnance is desired.

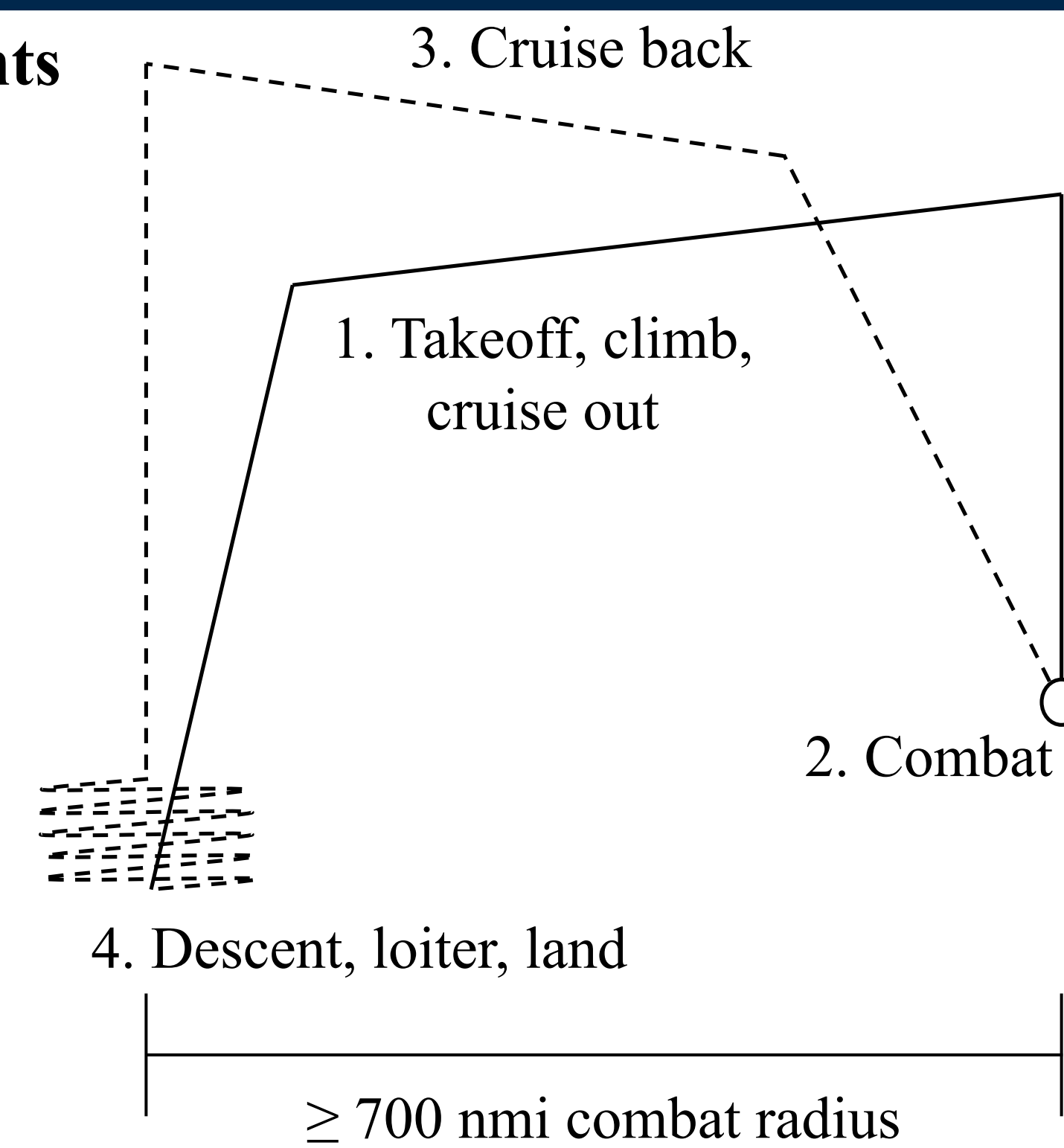
Air-to-Air Combat Mission

Minimum Performance Requirements

- 700 nmi combat radius
- 2 min combat at maximum thrust (10k ft)
- 8 deg/s sustained turn rate (20k ft)
- Mach 1.6 dash speed (30k ft)
- 7g vertical load factor (mid mission weight)

Payload

- 6x AIM-120C
- 2x AIM-9X



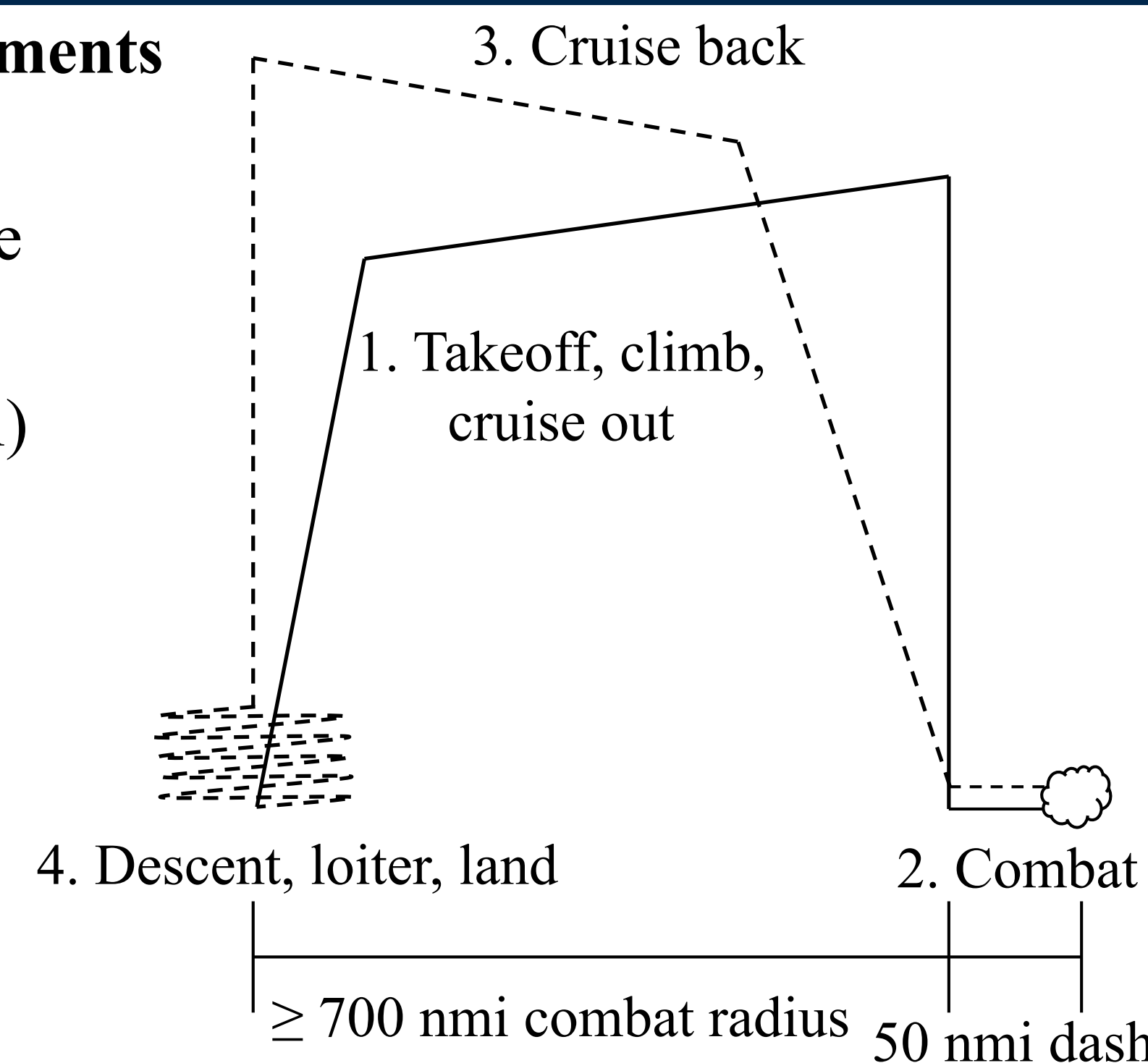
Strike Mission

Minimum Performance Requirements

- 700 nmi combat radius
- 2x 50 nmi dashes at intermediate thrust (sea level)
- Mach 0.85 dash speed (sea level)
- 7g vertical load factor (mid mission weight)

Payload

- 4x MK-83 JDAM
- 2x AIM-9X



Cost

Unit Flyaway Cost:

98.8 \$M

DOC/pay-nmi:

0.0138 \$

Total Engine Price:

18.8 \$M

Unit Flyaway Cost Cheaper than F/A-18E:

98.8 \$M < 113 \$M

*All costs reported in 2025 USD

Component	Cost Per Aircraft Per Year (\$)
Fuel	1,030,000
Oil	5,140
Aircraft Maintenance	2,050,000
Engine Maintenance	729,000
Crew	1,110,000
Surface Treatments	203,000
Miscellaneous	1,620,000
Total DOC	6,740,000



Key Geometric and Performance Parameters

Parameter	Value	Parameter	Value
Empty Weight (lb)	31,207	Cruise C_L	0.79
Takeoff Weight (lb)	60,826	Landing C_L	1.11
Air-to-Air Payload (lb)	2,470	Takeoff C_L	1.42
Strike Payload (lb)	4,424	Cruise L/D	11.7
Combat Radius (nmi)	1,000	Span (ft)	48.80
Air-to-Air Fuel Burn (lb)	26,959	Reference Area (ft ²)	647.80
Strike Fuel Burn (lb)	21,425	Aspect Ratio	3.67
T/W	0.98	Spot Factor	1.11
W/S (lb/ft ²)	94	Cruise Mach (max/economic)	1.8/0.84
Hookload (lb)	145,000	Engaging Speed (kts)	134

Empty Mass Budget

All-else empty:

- Avionics, flight and engine controls
- Instruments, hydraulics and electrical
- Furnishings, Aircon
- Handling Gear
- Fuel Systems, Inlet

Component	Weight (lb)	Component	Weight (lb)
Fuselage	5,018.2	Nose Gear	626.1
Wing	4,774.9	Main Gear	1,044.8
Horizontal Stabilator	510.5	Installed Engine	8,941.5
Vertical Tail	1,173.7	All-Else Empty	9,117.8

Key GFE

- AN/APG-81 Radar
- JPALS Approach & Landing System
- Quadruple redundant fly-by-wire
- Advanced Mission Computer
- Advanced Electronic Warfare system
- Life Support
 - Helmet, Ejection Seat, Oxygen

Propulsion System

2x F110-GE-132 Engines

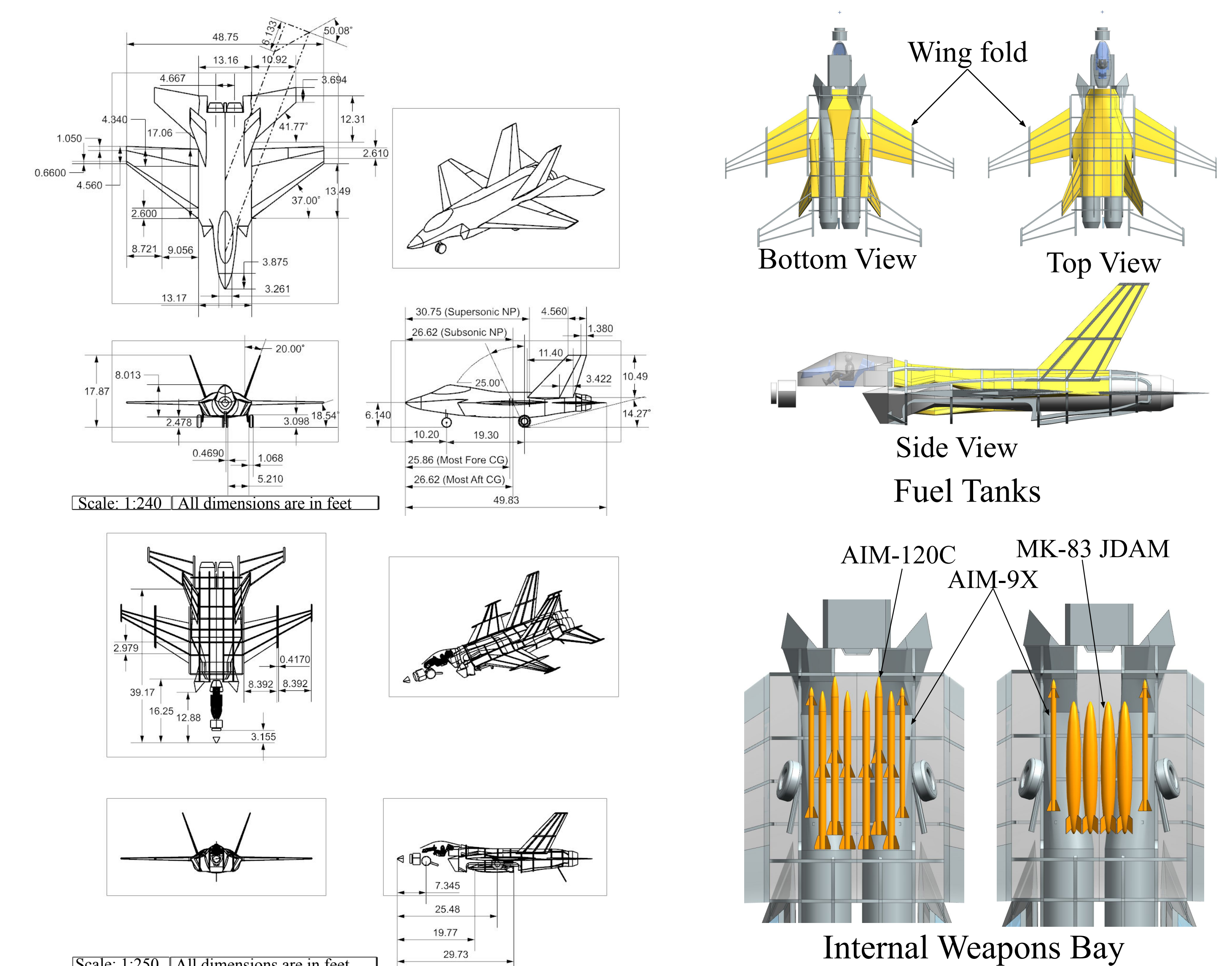
(maximum/military)

Thrust (SLS-lb): 33,093/17,669

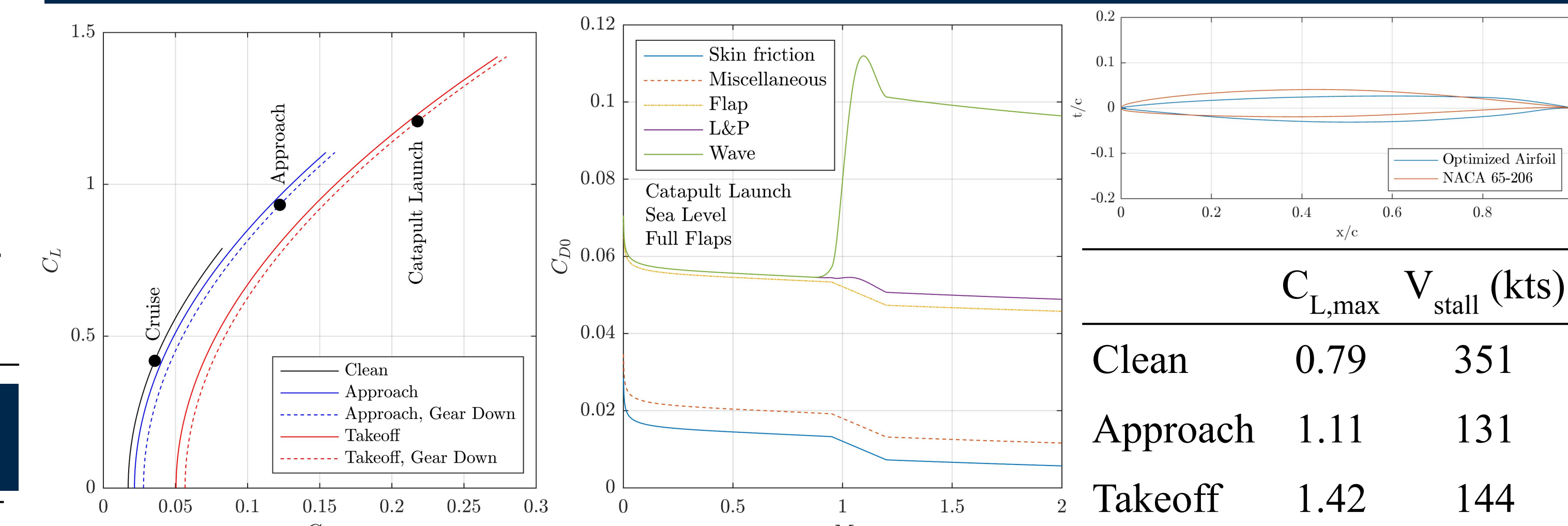
TSFC (lb/lbm-s): 0.68/1.90



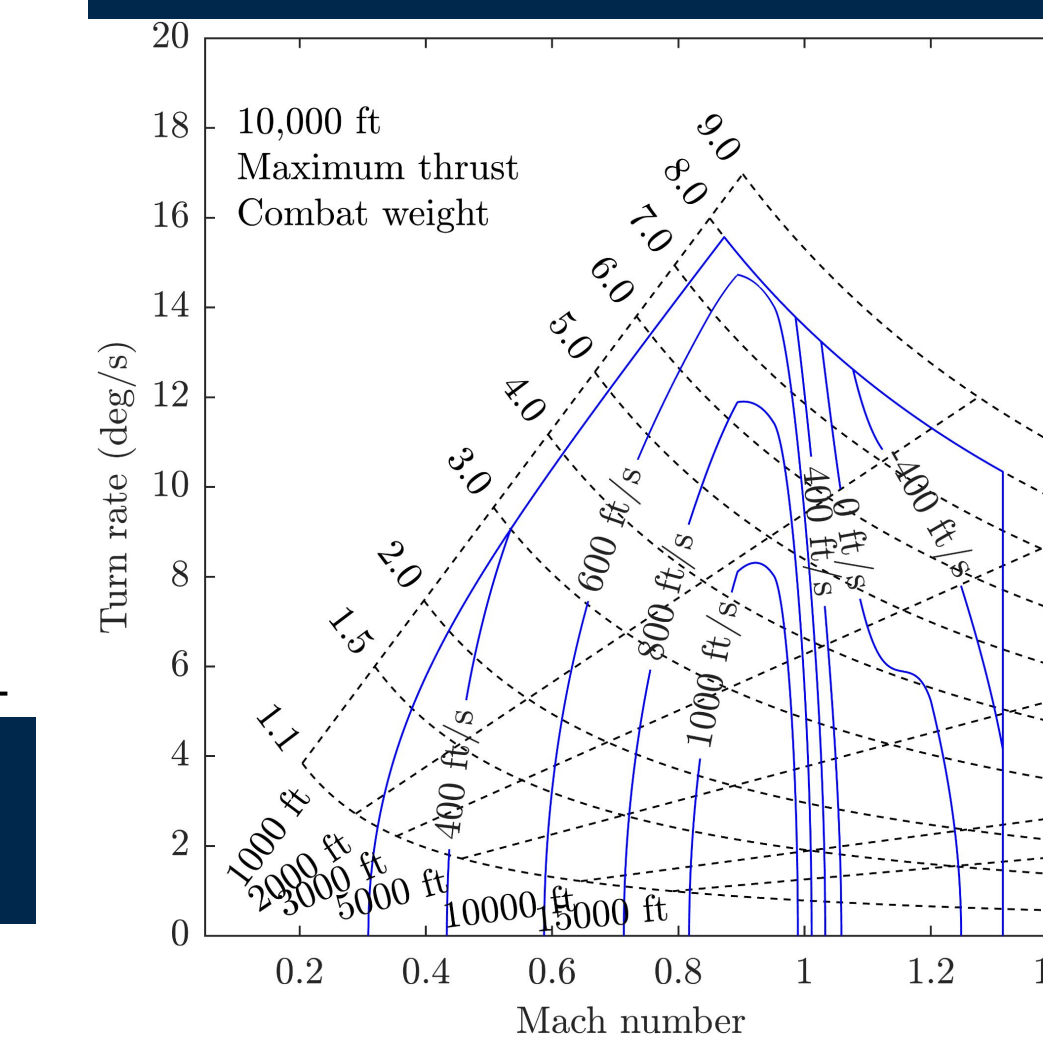
Dimensioned Drawings and Internal Layout



Aerodynamics

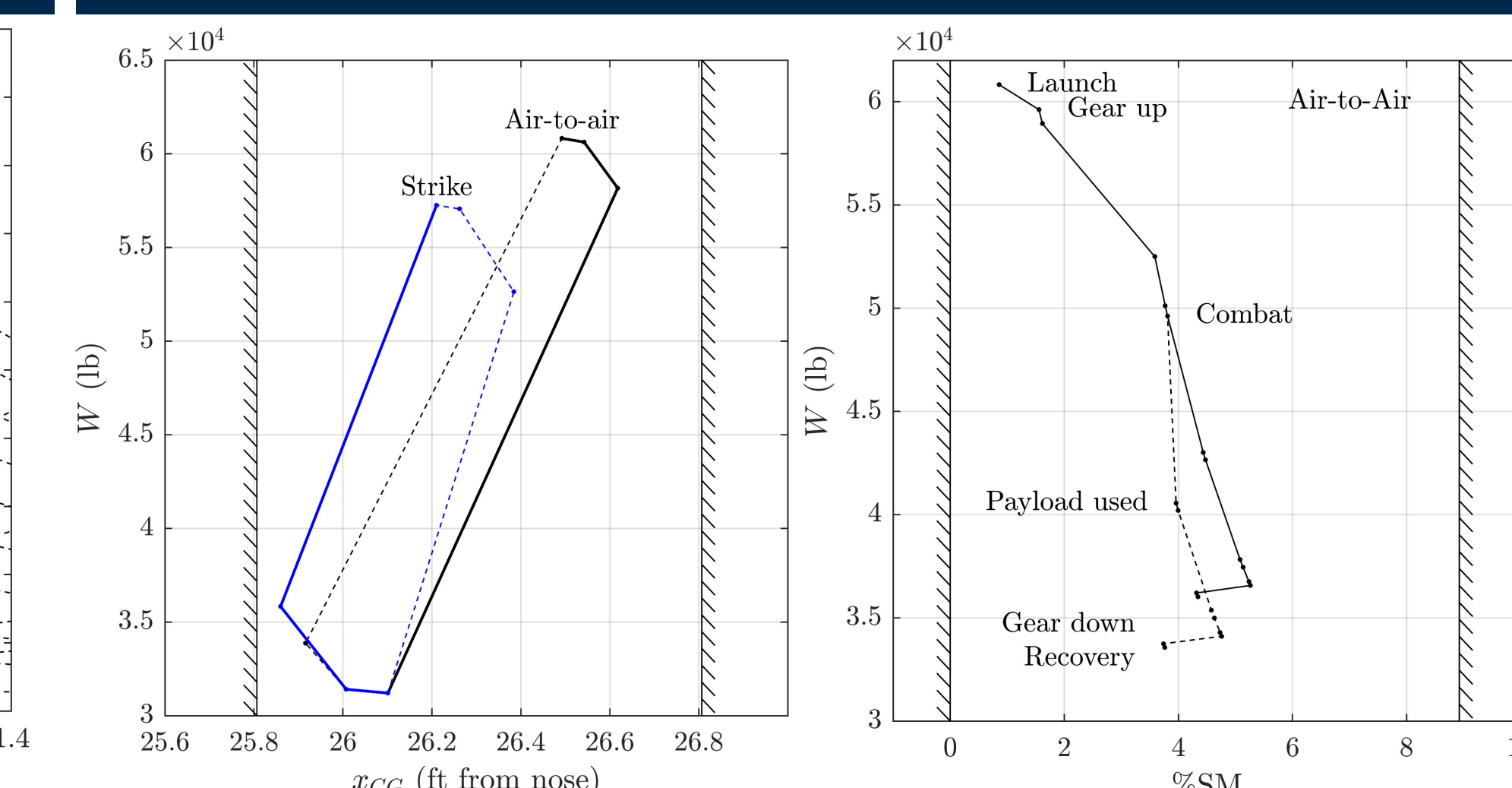


Performance



Energy-Maneuverability

Stability and CG



CG Loading

SM Envelope

References

- [1] Standard Aircraft Characteristics F/A-18E Super Hornet. NAVAIR00-110AF18-6. Boeing. 2001
- [2] Joaquim R.R.A. Martins. *The Metabook of Aircraft Design*. Dr. Martins, Joaquim R.R.A., 2021.
- [3] Leland M. Nicolai. *Fundamentals of Aircraft Design*. American Institute of Aeronautics and Astronautics, 2010. isbn: 978-1-60086-751-4.
- [4] Daniel P. Raymer. *Aircraft Design: A Conceptual Approach*. American Institute of Aeronautics and Astronautics, 2018. isbn: 9781624104909
- [5] Jan Dr. Roskam. *Airplane Design: A Conceptual Approach*. Roskam Aviation and Engineering Corporation, 1985.
- [6] F/A-18 corrosion maintenance doesn't consistently meet Navy and Marine Corps standards. 2021.
- [7] MCDONNELL DOUGLAS F/A-18 HORNET. 2018. url: <https://janes.migavia.com/usa/mde/f-18.html>.
- [8] List of military electronics of the United States. 2025. url: https://en.wikipedia.org/wiki/List_of_military_electronics_of_the_United_States.
- [9] AN/APG-81. 2016. url: <https://www.deagel.com/Components/ANAPG-81/a001381>.